

X problem

Tuesday, June 30, 2026 5:38 AM

Given the boundaries of 2 intervals. Print the boundaries of their **intersection**.

Note: **Boundaries** mean the two ends of an interval which are the starting number and the ending number.

Input

Only one line contains two intervals , where , .

It's guaranteed that and .

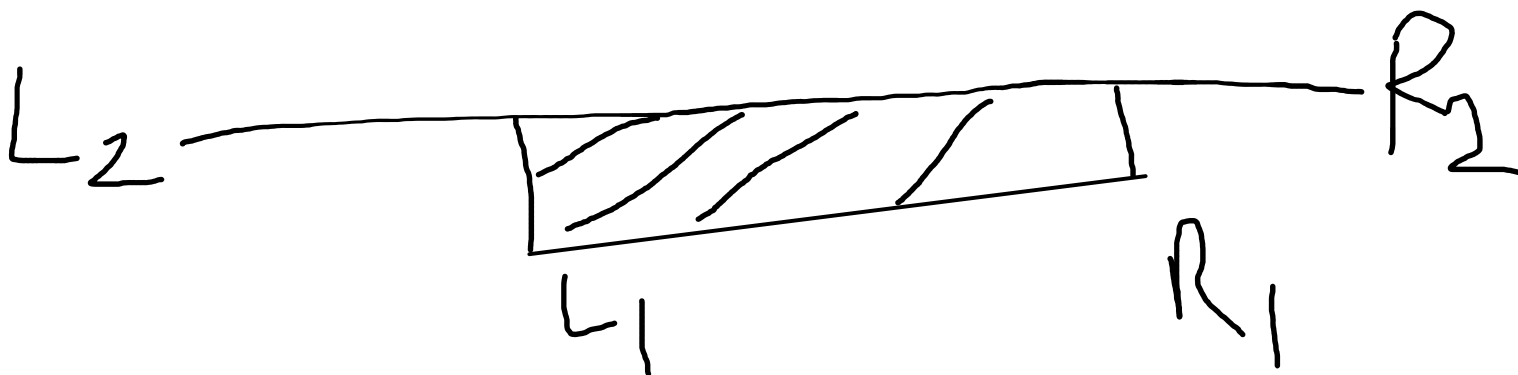
Output

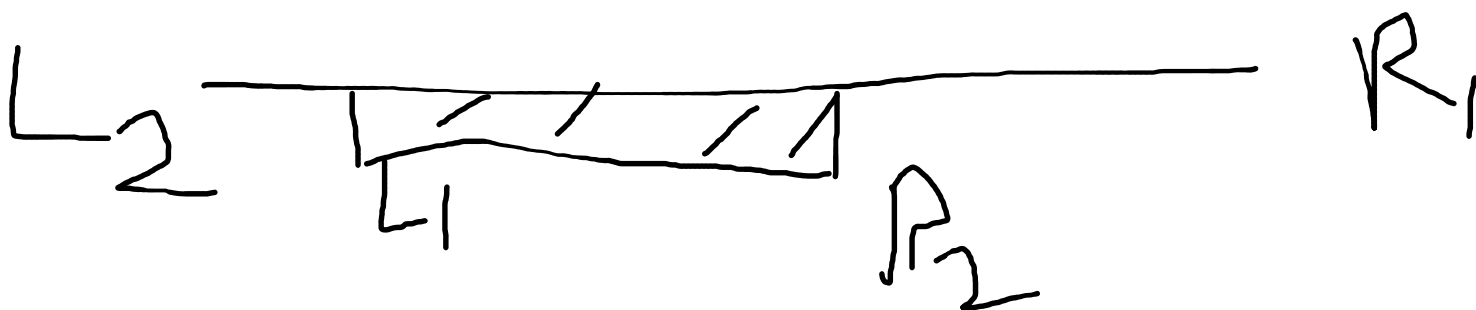
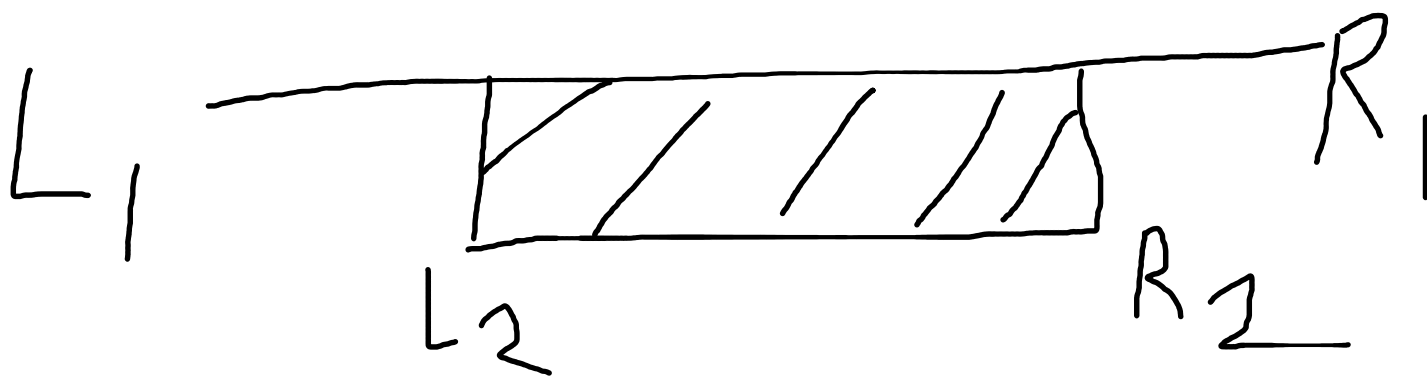
If there is an **intersection** between these 2 intervals print its boundaries , otherwise print -1.

From <https://codeforces.com/group/MWSDmaGsZm/contest/219158/problem/X>



intersect
At(L_2, R_1)





Modulus Operations

$$\downarrow$$

$$\frac{(a+b) \% m}{\uparrow}$$

$$\sum_{i=0}^n a_i$$

for $0 \rightarrow n$

$$\text{sum} += a$$

$$\text{sum} = \text{sum} + a$$

$$10^9 \neq 7$$

← Overflow

$$\underline{a + b = n}$$

→

$$(a \% m + b \% m) \% m = n \quad \text{not good}$$

$$\underline{a * b = n}$$

→

$$(a \% m * b \% m) \% m = n$$

$$\underline{a - b = n}$$

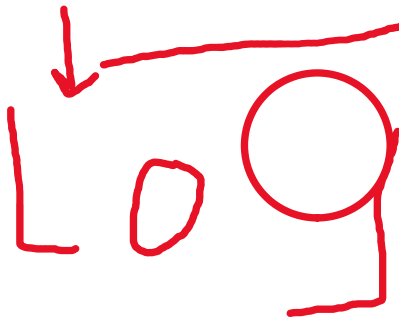
→

$$(a \% m - b \% m + m) \% m = n$$

$$a \% b \Rightarrow 0 \text{ to } b-1$$



long double > long long



if data
can't
carry

Large
No

Logarithm Rules

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Rules	Mathamatical Form
Product Rule	$\log_b(xy) = \log_b x + \log_b y$
Quotient Rule	$\log_b\left(\frac{x}{y}\right) = \log_b x - \log_b y$
Power Rule	$\log_b(x^n) = n \log_b x$
Change of Base Rule	$\log_b x = \frac{\log_c x}{\log_c b}$ or, $\log_b x \cdot \log_c b = \log_c x$
Zero Rule	$\log_b(1) = 0$
Identity Rule	$\log_b(b) = 1$
Equality Rule	$\log_b x = \log_b y \Rightarrow x = y$
Inverse Rule	$b^{\log_b x} = x \log_b(b^x) = x$
Reciprocal Rule	$\log_b\left(\frac{1}{x}\right) = -\log_b(x)$

Log Rules Chart

The following rules apply provided that...

$b \neq 1$ and $b > 0$, and M , N , and x are real numbers where $M > 0$ and $N > 0$

Rule	Property
Product Rule	$\log_b(MN) = \log_b M + \log_b N$
Quotient Rule	$\log_b\left(\frac{M}{N}\right) = \log_b M - \log_b N$
Power Rule	$\log_b(M^x) = x \cdot \log_b M$
Zero Rule	$\log_b(1) = 0$
Identity Rule	$\log_b(b) = 1$
Inverse Property of Logs	$\log_b(b^x) = x$
Inverse Property of Exponents	$b^{\log_b(x)} = x$
The Change of Base Formula	$\log_b a = \frac{\log_c a}{\log_c b}$

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